

In Silico Single-Shot Vacuum-Ultraviolet Pulse Reconstruction Via Solid-State High-Harmonic Generation

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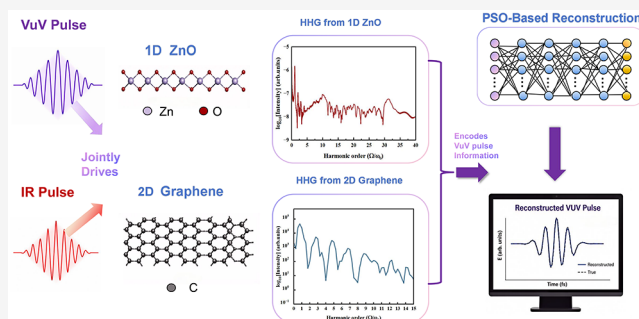


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Supporting Information

ABSTRACT: Vacuum-ultraviolet (VUV) pulses enable the initiation and control of coherent electronic dynamics on attosecond time scales, but phase-resolved characterization of VUV waveforms remains challenging due to strong absorption, limited nonlinear media, and shot-to-shot instabilities. Here we introduce an all-optical, single-shot metrology scheme based on solid-state high-harmonic generation (HHG). The method exploits laser-driven interband coherence in a semiconductor to encode the temporal amplitude and phase of an incident VUV field into the emitted harmonic spectrum. By combining a semiconductor Bloch equation or time-dependent Schrödinger equation simulations with a data-driven reconstruction framework, we demonstrate accurate retrieval of complex VUV waveforms in silico without delay scanning or multishot averaging. Interestingly, strong VUV absorption acts as an effective temporal gate that enhances phase sensitivity. These results establish solid-state HHG as a robust phase-sensitive probe, offering a practical route toward attosecond VUV metrology.



Nonlinear light–matter interactions in solids and molecules have emerged as a versatile platform for attosecond optics, ultrafast spectroscopy, and quantum control.¹ In particular, VUV pump–probe techniques provide direct access to electronic excitations resonant with bound and continuum states, enabling time-resolved studies of photoinduced chemical reactions, charge migration, and ultrafast modulation of optical properties in atoms, molecules, and condensed matter systems.^{2–4} Beyond these fundamental studies of attosecond electron dynamics, accurate single-shot reconstruction of VUV pulses is crucial for chemical and materials research. In molecular systems, phase-resolved VUV waveforms allow precise probing and control of ultrafast electronic excitations, facilitating investigations of conical intersections, nonadiabatic dynamics, and real-time charge redistribution.⁵ In condensed matter systems, VUV metrology enables selective excitation of specific electronic bands or localized states in semiconductors, low-dimensional materials, and nanostructures, supporting time-resolved studies of carrier injection, exciton formation, and light-induced phase transitions.^{2,6,7}

Despite sustained experimental and theoretical efforts, reliable temporal characterization of ultrashort VUV pulses remains challenging, as explicitly noted by,⁷ “the last—and inevitably most difficult—challenge is the full temporal characterization of the broadband few-cycle ultraviolet pulses”.

These challenges arise from both the experimental and theoretical aspects. Experimentally, strong absorption and dispersion in the VUV regime impose stringent requirements on vacuum conditions and beam transport, while the lack of

broadband, low-loss nonlinear media severely restricts conventional optical characterization techniques.^{3,7,8}

In addition, traditional strong field approximation and second-order perturbation theory-based methods such as attosecond streaking cameras⁹ and frequency-resolved optical gating¹⁰ rely on multishot measurements, in which the temporal waveform is reconstructed by accumulating signals over many laser shots while scanning an external parameter such as delay or phase. In the VUV regime, however, pulse-to-pulse fluctuations in carrier-envelope phase, intensity, and timing jitter are difficult to fully stabilize. As a result, the reconstructed waveform often represents an averaged response rather than the true single-shot pulse structure. These limitations motivate the development of single-shot, all-optical approaches capable of retrieving attosecond VUV waveforms without parameter scanning or multishot averaging.^{2,6–8}

Furthermore, an additional difficulty lies in accurately modeling the interaction between the VUV pulse and the target within the reconstruction framework: VUV pulse reconstruction schemes based on single- or two-photon ionization spectra of low-ionization materials (such as metals)

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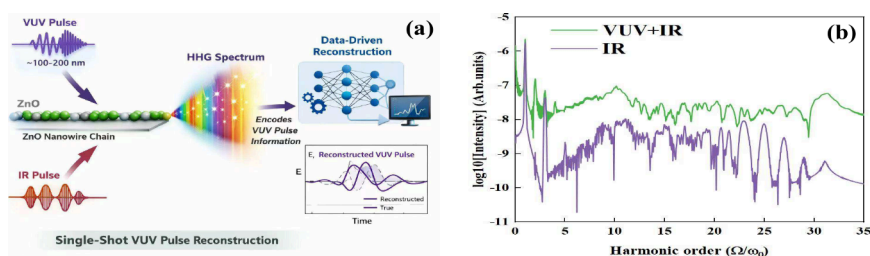


Figure 1. (a) Schematic of the reconstruction framework: An IR driving field and a VUV probe field are copropagated onto a ZnO nanowire medium. Through solid-state HHG, the medium emits harmonic spectra that encode the temporal characteristics of the incident VUV pulse. A data-driven machine learning model is then used to invert the harmonic spectral information and reconstruct the complete temporal waveform of the VUV probe field. (b) Logarithmic intensity plot of HHG spectra from ZnO under two excitation conditions: the green curve corresponds to HHG driven by the combined IR+VUV fields, while the purple curve corresponds to HHG driven by the IR field alone. A significant enhancement in harmonic intensity is observed in the spectral plateau region with the VUV probe field, demonstrating the pronounced modulation of the medium's nonlinear response by the VUV field.

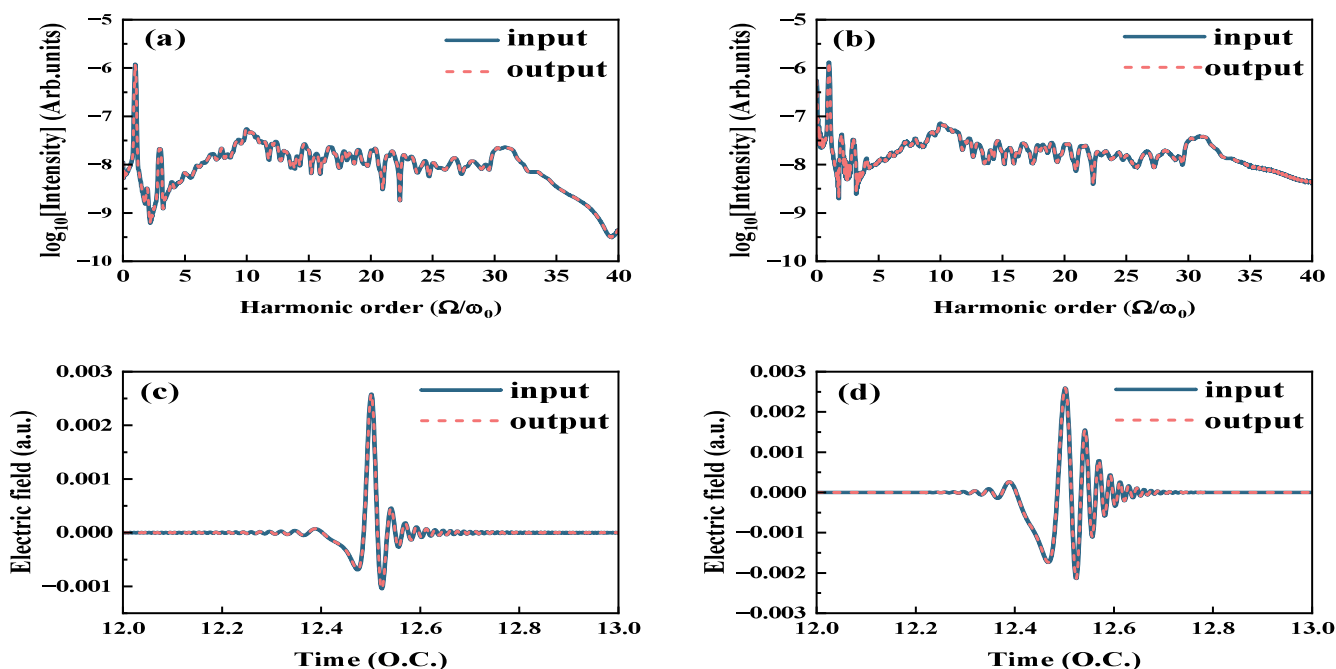


Figure 2. Blue solid line represents the input value (original spectrum or pulse), and the pink dashed line represents the value reconstructed via the fitting process (reconstructed spectrum or pulse). (a) and (c) display the results for the VUV electric field with a Lorentzian line shape: (a) shows the fitted curve of the HHG spectrum, and (c) shows the corresponding fitted curve of the VUV pulse waveform, with a fitting error of 8.836×10^{-8} . (b) and (d) present the results for the VUV electric field envelope with a sech^2 (hyperbolic secant) functional form: (b) shows the fitted curve of the HHG spectrum, and (d) shows the corresponding fitted curve of the VUV pulse waveform, with a fitting error of 4.317×10^{-7} .

are intrinsically affected by electron–nuclear Coulomb interactions and strong material absorption, both of which introduce non-negligible distortions to the measured spectra.^{6–8} These effects are not rigorously accounted for within conventional semianalytical theory frameworks. Consequently, widely employed techniques such as attosecond streaking cameras and frequency-resolved optical gating may suffer from systematic inaccuracies when applied in the VUV regime.

To overcome these limitations, we introduce a solid-state platform for single-shot reconstruction of attosecond VUV pulses based on the intrinsic phase sensitivity of solid-state HHG. Since the first observation of HHG in bulk crystals, solid-state HHG has emerged as a promising avenue for ultrafast metrology due to its unique electronic response mechanisms.^{4,11} Unlike conventional schemes that rely on external nonlinear or ionization media, solid-state HHG provides an all-optical response in which the temporal

information on the incident field is encoded through coherent interference between the intrinsic nonlinear polarization driven by the IR field and the polarization perturbed by the VUV excitation. Importantly, strong absorption in the VUV range, typically regarded as a major obstacle for pulse characterization,^{2,3,6–8} does not degrade the reconstruction fidelity in the present scheme but instead enhances the sensitivity of the harmonic response by confining the interaction to a well-defined microscopic region.

Figure 1 (a) illustrates the basic concept of the proposed scheme. A VUV pulse and an infrared (IR) driving pulse simultaneously interact with a ZnO nanowire, with their relative temporal delay fixed at zero, defined as the temporal overlap of the peak intensities of the two pulses. This joint interaction generates solid-state high-harmonic radiation, in which the temporal information on the incident VUV pulse is imprinted onto the resulting HHG spectrum. Figure 1(b)

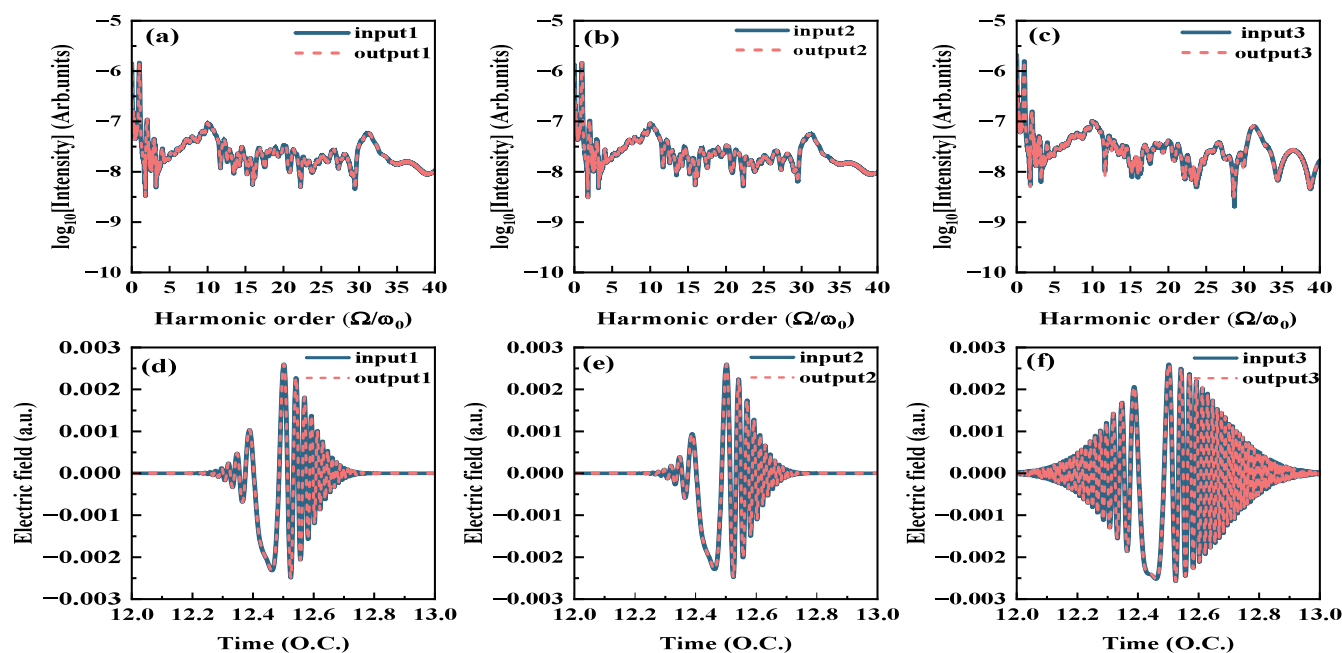


Figure 3. (a) and (d) serve as the reference group, where the VUV pulse has a fwhm of 1000 as. (b) and (e) constitute the narrowed group, where the VUV pulse fwhm is adjusted to 950 as while keeping all other reference conditions unchanged. (c) and (f) form the broadened group, where the VUV pulse fwhm is adjusted to 2000 as while maintaining other reference conditions constant. The corresponding results for each group are as follows: In each group, (a)/(b)/(c) display the corresponding HHG spectra, and (d)/(e)/(f) show the corresponding reconstructed VUV pulse waveforms. The fitting errors for the narrowed group and the broadened group are 2.412×10^{-7} and 8.198×10^{-5} , respectively.

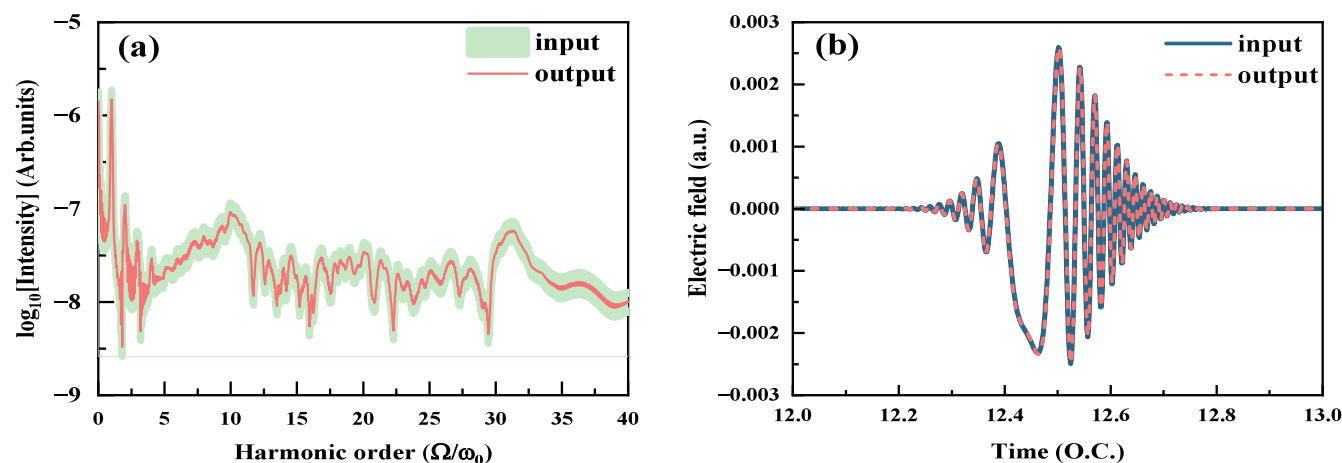


Figure 4. To test the robustness of the algorithm, a random intensity error of $\pm 10\%$ was introduced into the input high-harmonic signal (i.e., the harmonic spectrum in the presence of the VUV field, as shown in Figure 1(b)). In the figure, the pink curve and the green squares represent the input noisy harmonic spectrum and its fitted result, respectively. In panel (b), the blue solid line and the pink dashed line denote the input and the fitted-and-reconstructed VUV electric field, respectively. The final fitting error is 0.002012.

compares the HHG spectra generated by the IR field alone and by the combined VUV+IR fields with semiconductor Bloch equations (SBE),^{12–14} which capture interband polarization and intraband carrier motion within a band-structure description. A pronounced modification of the harmonic response is observed in the presence of the VUV pulse, indicating a strong sensitivity of solid-state HHG to the temporal structure of the incident VUV field. The encoded VUV waveform is subsequently retrieved using the data-driven reconstruction algorithm developed in our previous work.^{15,16}

Within the reconstruction framework, the solid-state HHG response is obtained through a self-consistent numerical treatment of laser-driven electron dynamics. Specifically, we

solve either the two-band SBE or the time-dependent Schrödinger equation (TDSE) in reciprocal space under a periodic potential.^{17,18} The latter formulation naturally incorporates multiband coupling effects. This dual-model strategy ensures that the reconstructed VUV waveform is not an artifact of a particular theoretical approximation but instead reflects the intrinsic nonlinear response of the solid-state system. Detailed formulations of the SBE and the TDSE are referred to the relevant literature,^{17,19} and the detailed formulation of the data-driven reconstruction algorithm is provided in the [Supporting Information](#).

The pronounced sensitivity of solid-state HHG to the incident VUV pulse originates from the coherent coupling

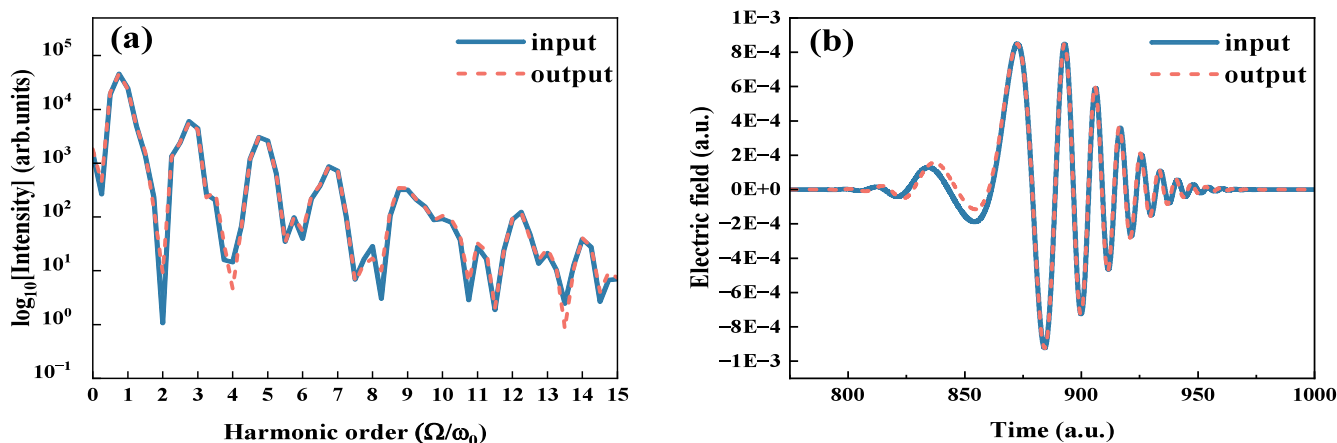


Figure 5. (a) Reconstruction results of HHG spectra for the 2D graphene system: the blue curve represents the input HHG spectrum, and the pink curve represents the reconstructed HHG spectrum output by the data-driven model. (b) Temporal waveform of the VUV electric field under the corresponding conditions: the blue curve represents the input temporal waveform of the VUV pulse, and the pink curve represents the reconstructed temporal waveform of the VUV pulse. The overall fitting error is 0.05963, indicating that the reconstructed results are highly consistent with the characteristics of the original signals, and the data-driven method exhibits excellent accuracy and reliability in both HHG spectrum and temporal pulse reconstruction.

between laser-driven interband polarization, intraband current, and the band-resolved electronic structure of the solid. Unlike gas-phase HHG, which relies on electron ionization, continuum propagation, and recombination under long-range Coulomb interaction, solid HHG is governed by electron–hole dynamics confined within a finite band structure.^{4,20} As a result, even a weak VUV field can efficiently perturb the initial interband coherence and population distribution, leading to a measurable modification of the subsequent harmonic emission driven by the infrared field.

Moreover, strong material absorption in the VUV range selectively couples the incident pulse to specific interband transitions, imprinting the temporal phase and amplitude of the VUV field onto the microscopic polarization with high fidelity.²¹ This selective excitation enhances the contrast between different temporal features of the VUV waveform in the emitted harmonic spectrum, thereby improving reconstruction accuracy. In this pump–probe configuration, the VUV pulse acts primarily as an ultrafast preparation field rather than a primary energy source, relaxing the requirement on photon flux and rendering solid-state HHG particularly suitable for single-shot characterization of low-flux VUV pulses generated by HHG-based sources.²² Notably, this potential of solid HHG as an intrinsic, phase-sensitive probe has remained largely unexplored so far.

Figures 2, 3, 5, and 4 present representative reconstruction benchmarks of the proposed scheme. All laser parameters, reconstruction accuracy, and computational time for each case are summarized in Table S1 of Supporting Information. Figures 2, 3, and 4 present HHG spectra calculated within a one-dimensional (1D), two-band SBE framework, whereas Figure 5 employs a 2D TDSE approach to model solid-state HHG in a periodic potential.

Within the SBE simulations, the following approximations are adopted: (i) Only one valence band and one conduction band are assumed to participate in the laser–matter interaction, while contributions from higher-lying bands are neglected. This approximation is expected to capture the dominant carrier dynamics in the moderate-field regime considered here. (ii) To reduce computational cost while

preserving experimental relevance, a 1D ZnO nanowire is employed as the model system. Such ZnO nanostructures have been experimentally realized, supporting the practical feasibility of the proposed scheme.^{11,23}

For the 2D TDSE simulations of HHG in crystalline materials,¹⁷ the single-electron approximation and the electric-dipole approximation are adopted.

In both theoretical frameworks, during the reconstruction procedure, the temporal waveform of the VUV pulse,

$$E_X(t) = f(t)\cos[\Omega t + g(t)]$$

is treated as the sole unknown quantity to be retrieved, while all other parameters, including the IR driving field and the electronic properties of ZnO, are assumed to be known *a priori*. The impact of uncertainties in these parameters is discussed in the Supporting Information.

It should be emphasized that the “single-shot” reconstruction presented here refers to the use of a single simulated HHG spectrum as the sole input for waveform retrieval, establishing theoretical feasibility. Experimental realization will require additional considerations such as detector noise, calibration, and stability, which are partially addressed by our noise robustness tests.

In Figure 2, we consider two VUV probe pulses with distinct temporal waveforms (blue solid lines in Figures 2(c) and 2(d)). In both cases, the reconstructed pulses accurately reproduce the target waveforms, demonstrating the ability of the method to retrieve the complex temporal structures.

We next assess the performance of the reconstruction for VUV pulses with different full widths at half-maximum (fwhm). Although previous studies have reported the generation of VUV pulses as short as 950 as in ZnO,²⁴ femtosecond and sub-femtosecond VUV pulses remain highly desirable for practical applications. As shown in Figure 3, we test reconstruction examples for VUV pulses with durations of 950 as, 1 fs, and 2 fs. The results indicate that the proposed method maintains high reconstruction fidelity across both the attosecond and femtosecond regimes.

The robustness of the reconstruction against measurement noise is a critical consideration for experimental feasibility. To

emulate realistic experimental conditions, random noise with a maximum amplitude of 10% was added to each HHG spectrum, and the noisy spectra were used as inputs for the reconstruction. As shown in Figure 4, the method retains good performance even in the presence of 10% noise, highlighting its strong noise tolerance.

All of the preceding examples are based on a 1D, two-band SBE framework. To extend the method to multidimensional and multiband crystalline systems, we further demonstrate an alternative reconstruction strategy based on a single-electron two-dimensional TDSE approach. Specifically, HHG spectra are calculated in a periodic potential using our open-source 3D-GTDSE solver.²⁵ This framework has been extensively adopted in the theoretical description and simulation of solid-state HHG.¹⁴ Compared with the SBE framework, the TDSE approach naturally incorporates multiband and higher-dimension dynamics with lower calculation cost and is therefore well-suited for reconstructing VUV pulses with higher central photon energies.^{2,17,26,27}

However, unlike the SBE formalism, which implicitly accounts for multielectron effects through band dispersion and interband transition dipole moments, the single-electron TDSE neglects explicit electron–electron interactions, although the role of multielectron effects in solid-state HHG remains under active debate in the literature, with contrasting conclusions reported. Here, we choose a 2D hexagonal graphene system as the HHG medium, with both the VUV and infrared fields polarized in the plane of the material. Apart from the dimensionality and target material, the reconstruction procedure remains identical to that used in Figures 2–4. As shown in Figure 5, excellent agreement between the input and reconstructed VUV pulses is obtained, confirming that the proposed method is applicable to multidimensional and multiband solid-state systems.

Developing a unified reconstruction framework that combines the complementary advantages of the SBE and TDSE formalisms represents a key objective for our future research.

We have also verified the uniqueness and robustness of our reconstruction by testing multiple random initial guesses and different IR field parameters, all converging to the same accurate waveform (See Supporting Information Figures S1–S2). These results, together with the temporal gating effect of strong VUV absorption, indicate that the mapping is practically unique and the loss landscape is sufficiently convex.

Our simulations neglect phenomenological dephasing and explicit electron–electron interactions. In real materials, dephasing occurs on a few-to-tens-of-fs time scale; for attosecond to few-fs VUV pulses (shorter than this time scale), the encoded phase information is preserved. Strong dephasing (e.g., in defective samples) would reduce modulation depth and reconstruction fidelity, whereas electron–electron interactions in wide-band gap semiconductors primarily renormalize band parameters rather than destroy the phase mapping. Our 10% noise-robustness tests already implicitly account for moderate experimental imperfections.

In conclusion, we have proposed a solid-state HHG-based platform for the single-shot, phase-resolved reconstruction of attosecond VUV pulses. By encoding the temporal information on the incident VUV field into a single harmonic spectrum, the method enables all-optical pulse characterization without delay scanning or multishot averaging. The robustness of the reconstruction against noise, pulse duration, waveform

complexity, and theoretical descriptions indicates that the underlying principle is governed by generic features of laser-driven band dynamics rather than material-specific details.

The efficacy of the reconstruction scheme originates from the microscopic coupling between VUV-induced electronic coherence and strong-field-driven carrier dynamics. In solid-state HHG, the emitted spectrum arises from a coherent superposition of interband polarization contributions.²⁰ When a VUV pulse interacts with the semiconductor, it prepares an initial coherent superposition of valence and conduction band states, imprinting its temporal phase and amplitude onto the electron–hole wave packets.^{4,28} The subsequent IR-driven intraband acceleration nonlinearly maps this initial phase information onto the emitted harmonic radiation.²⁹ The resulting harmonic spectrum reflects the interference between the intrinsic polarization driven by the IR field and the polarization perturbed by the VUV excitation.³⁰ Owing to the subcycle sensitivity of carrier injection in solids, even subtle variations in the VUV waveform lead to measurable redistributions of the harmonic yield. In this effective sense, the solid-state medium acts as a self-referencing attosecond interferometer, in which the electronic band structure provides the nonlinear mapping required to encode the complete temporal information on the VUV field.³

Future extensions of this work could leverage machine learning algorithms^{31–35} to enhance reconstruction speed and accuracy and expand the scheme to other low-dimensional materials. This approach opens new avenues for attosecond metrology in condensed matter systems, with potential applications in ultrafast electron dynamics and quantum control.

■ ASSOCIATED CONTENT

Data Availability Statement

All the data and code in this work are available upon publication, the TDSE solver can be downloaded from the Science Data Bank at [10.57760/sciencedb.27107](https://doi.org/10.57760/sciencedb.27107), and the user guide can be found in ref.25 The optimization code is publicly available from the Supporting Information of ref.36

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpclett.6c00844>.

The influences of key IR laser parameters (field intensity, fwhm, and frequency) on VUV pulse reconstruction. The influence mechanism of IR laser intensity uncertainty on the accuracy of VUV pulse reconstruction. Detailed formulation of the data-driven reconstruction algorithm (PDF)

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Notes

The authors declare no competing financial interest.

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